MODELING OF FOOTBALL MATCHES BY THE POISSON LAW AND APPLICATION TO EURO 2024

SUMMARY

INTRODUCTION

01 BUILDING A SIMPLE MATHEMATICAL MODEL TO PREDICT THE SCORE OF A MATCH

02 SECOND MODEL: POISSON REGRESSION

03 IMPROVEMENT OF THE SECOND MODEL: TAKING INTO ACCOUNT THE FIFA RANKING

04 COMPARISON TO ACTUAL RESULTS: EURO 2020

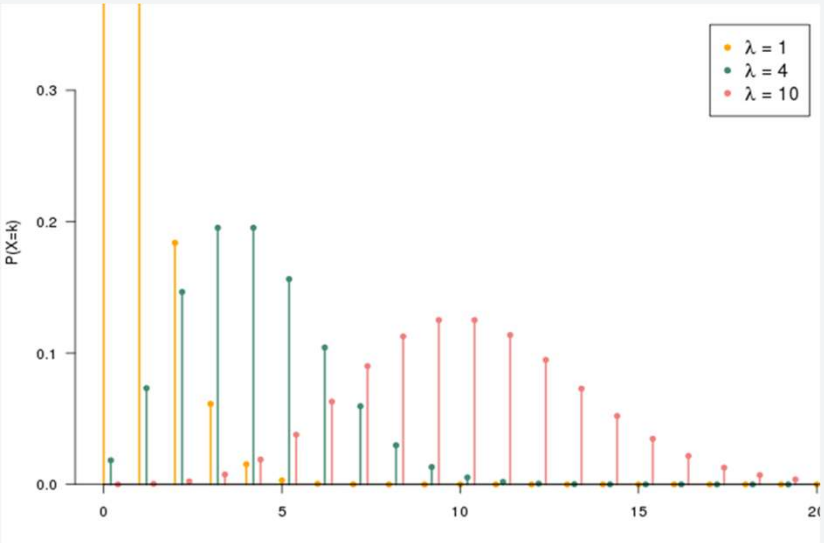
05 APPLICATION TO EURO 2024: PREDICTION OF RESULTS

CONCLUSION AND DISCUSSION

INTRODUCTION

• POISSON DISTRIBUTION FOR FOOTBALL MATCH MODELING

- **Discrete probability law**
- **Describes the behavior of the number of events in a fixed time interval, if those events occur with an average frequency**



Appearance of the Poisson distribution for several values of λ

- **If the average number of occurrences in a fixed time interval is λ, the probability that there are x occurrences is:**

$$P(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}, \lambda > 0$$

INTRODUCTION

- **COMPARISON OF THE POISSON DISTRIBUTION TO THE DISTRIBUTION OF GOALS OVER 5 SEASONS OF 2017-2022 IN THE 5 MAJOR LEAGUES**

Overall goal distribution over the 5 seasons

Poisson distribution - result distribution

Match Resu..	Distribution prediction distribution	real distribution
0-0	7,97%	7,15%
0-1	8,50%	7,38%
1-0	10,55%	9,32%
1-1	10,76%	11,78%
1-2	6,28%	6,67%
2-1	7,65%	8,86%
2-2	4,24%	5,05%
2-3	1,77%	1,97%
3-2	2,13%	2,33%
3-3	0,85%	1,05%
3-4	0,28%	0,26%
4-3	0,34%	0,34%
4-4	0,11%	0,09%
4-5	0,03%	0,04%
5-4	0,04%	0,03%
5-5	0,01%	0,00%

Comparison between the real distribution of scores and the Poisson distribution

INTRODUCTION

DATA AVAILABLE:

France National Team Data Download

Match Data	UEFA Euro Qualifiers 2022/2024 Matches CSV
	International Friendlies 2024/2024 Matches CSV
	UEFA Euro Championship 2024/2024 Matches CSV
	UEFA Nations League 2024/2025 Matches CSV

2022-2024 international matches, www.footystats.org

- **Euro 2024 qualifying matches: 2022-2024**
- **International friendlies: 2023-2024**
- **Nations League: 2022-2024**

	A	B	C	D
1	Equipe 1	Equipe 2	Buts equipe 1	Buts equipe 2
2	Italy	England	1	1
3	England	Denmark	2	1
4	Italy	Spain	1	1
5	Ukraine	England	0	4
6	Czech Republic	Denmark	1	2
7	Belgium	Italy	1	2
8	Switzerland	Spain	1	1
9	Sweden	Ukraine	1	2
10	England	Germany	2	0
11	France	Switzerland	3	3
12	Croatia	Spain	3	5
13	Belgium	Portugal	1	0
14	Netherlands	Czech Republic	0	2
15	Italy	Austria	2	1
16	Wales	Denmark	0	4
17	Portugal	France	2	2
18	Germany	Hungary	2	2
19	Sweden	Poland	3	2
20	Slovakia	Spain	0	5
21	Croatia	Scotland	3	1
22	Czech Republic	England	0	1
23	Finland	Belgium	0	2
24	Russia	Denmark	1	4
25	Ukraine	Austria	0	1
26	North Macedonia	Netherlands	0	3
27	Italy	Wales	1	0

Retrieving and formatting data in a spreadsheet program

INTRODUCTION

DATA TO ESTIMATE THE RELEVANCE OF THE MODEL: ODDS ON ODDSPORTAL

Euro 2024 Fixtures & Betting Odds

Next Matches | Outrights | Results | Standings

Football / Europe / Euro 2024

14 Jun 2024		1	X	2	B's
21:00	Germany - Scotland	1.28	5.86	10.99	14
15 Jun 2024		**1**	**X**	**2**	**B's**
15:00	Hungary - Switzerland	3.62	3.34	2.14	14
18:00	Spain - Croatia	1.78	3.65	4.78	14
21:00	Italy - Albania	1.37	4.75	9.42	14
16 Jun 2024		**1**	**X**	**2**	**B's**
15:00	Poland - Netherlands	5.09	3.98	1.67	14
18:00	Slovenia - Denmark	5.12	3.62	1.74	14
21:00	Serbia - England	6.87	4.43	1.48	14
17 Jun 2024		**1**	**X**	**2**	**B's**
15:00	Romania - Ukraine	3.65	3.46	2.08	14
18:00	Belgium - Slovakia	1.48	4.47	6.72	14
21:00	Austria - France	6.36	4.53	1.50	13

Upcoming match predictions by bookmakers, oddsportal

	Team1	Team2	prob1	probX	prob2
0	Hungary	Switzerland	0.304878	0.290698	0.446429
1	Spain	Croatia	0.578035	0.264550	0.201207
2	Italy	Albania	0.729927	0.211416	0.107875
3	Poland	Netherlands	0.196464	0.247525	0.598802
4	Slovenia	Denmark	0.211416	0.273973	0.558659
5	Serbia	England	0.145138	0.220264	0.680272
6	Romania	Ukraine	0.283286	0.297619	0.460829
7	Belgium	Slovakia	0.671141	0.222717	0.151057
8	Austria	France	0.145773	0.213220	0.684932
9	Turkey	Georgia	0.564972	0.270270	0.210526
10	Portugal	Czech Republic	0.632911	0.244499	0.163934

Data Formatting

01 - CONSTRUCTION OF A FIRST SIMPLE MODEL

MODEL

- **•λ represents the average number of goals scored by team 1 and μ by team 2**
- **• The number of goals scored by the home team and the opposing team is considered as two independent distributions of Poisson**

$$P(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}, \lambda > 0$$

Proba that team 1 scores x goals

$$P(y) = \frac{e^{-\mu}\mu^{y}}{y!}, \mu > 0$$

Proba that team 2 will score y goals

Appearance of the Poisson distribution for several values of λ

$$P(X_{i,j} = x, Y_{j,i} = y) = \frac{e^{-\lambda}\lambda^{x}}{x!}\frac{e^{-\mu}\mu^{y}}{y!}$$

CONSTRUCTION OF A SIMPLE FIRST MODEL

HOW TO CHOOSE Λ?

- **For each team, we define:**
- **-α: attacking strength = average of goals scored by the team over the matches in the dataset**
- **-β: Defensive strength = average goals conceded over matches in the dataset**

Equipe	Buts marqués	Buts concédés
Albania	1.500000	0.500000
Andorra	0.300000	2.000000
Armenia	1.125000	1.375000
Austria	2.125000	0.875000
Azerbaijan	0.875000	2.125000
Belarus	0.900000	1.400000
Belgium	2.750000	0.500000
Bosnia-Herzegovina	0.909091	2.000000
Bulgaria	0.875000	1.750000

Hence an intuitive modeling of λ and μ:

$$\lambda = \alpha_i \beta_j \qquad \mu = \alpha_j \beta_i$$

CONSTRUCTION OF A SIMPLE FIRST MODEL

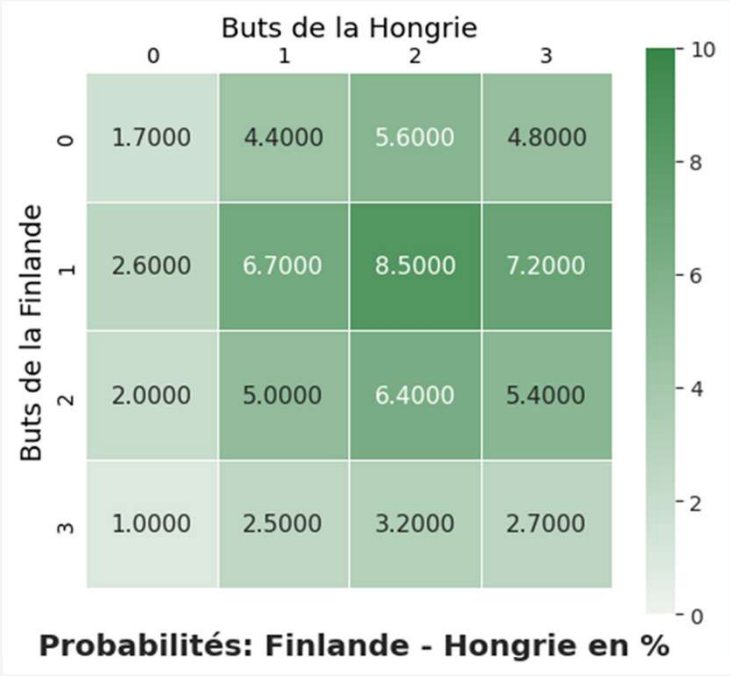
RESULT: 1ST PREDICTION OF FINLAND - HUNGARY

- **We limit ourselves to 3 goals for readability**
- **Most likely score: Finland 1 – Hungary 2**
- **Probability:**

$$P\left(X_{\text{Hongrie}} = 2, Y_{\text{Finlande}} = 1\right) = \frac{e^{-\lambda}\lambda^2}{2!}\frac{e^{-\mu}\mu^1}{1!}$$

=8.5%

tel que: $\lambda = \alpha_{\text{Hongrie}}\beta_{\text{Finlande}}$ $\mu = \alpha_{\text{Finlande}}\beta_{\text{Hongrie}}$



CONCLUSION

22%
- **Finland wins**

19%
- **Draw**

59%
- **Hungary wins**

CONSTRUCTION OF A SIMPLE FIRST MODEL

RELEVANCE OF THE MODEL:

Comparison of results to bookmakers' predictions

- **We transform the odds of sports betting sites into probabilities for each match not yet played (June 2024 Calendar)**
- **We calculate the difference in probability of each outcome between our model and the bookmakers' predictions**

D'où la différence de prévision en moyenne pour le 1er modèle:

Team 1 victory
17% difference

Equality
9% difference

Team 2 Victory
16% difference

02 - POISSON REGRESSION

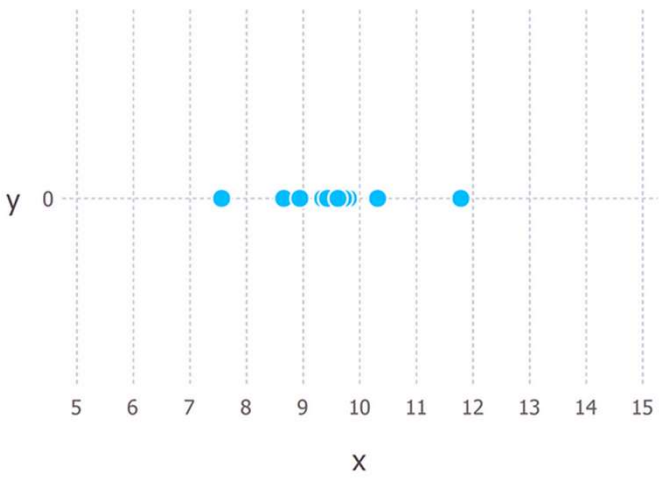
•A- LIKELIHOOD FUNCTION

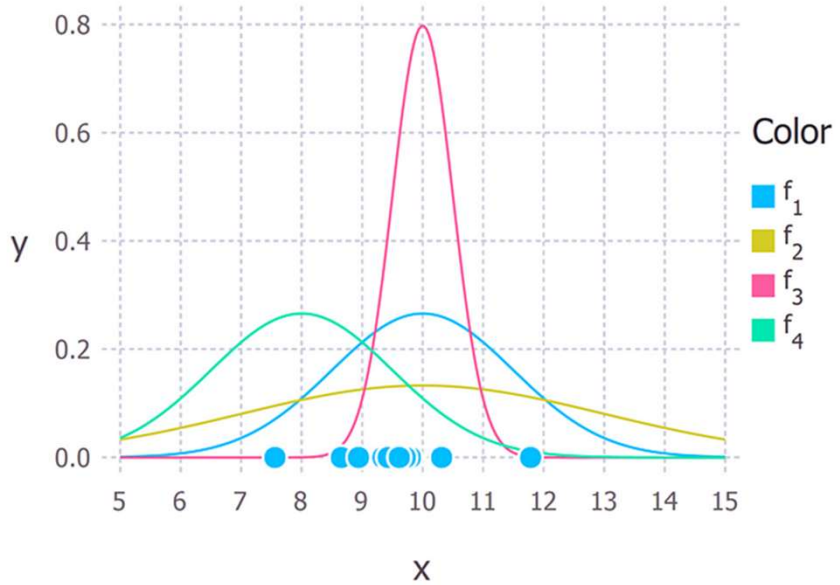
- **•Objective: To determine precisely the most suitable λ for our dataset for each team.**

Soit $X_1, X_2, ..., X_n$ une suite de variables aléatoires i.i.d de paramètres θ, où $\theta \epsilon \mathbb{R}^p$.

$$X_1 = x_1, X_2 = x_2, ..., X_n = x_n$$

- **It is assumed to have observed:**





RÉGRESSION DE POISSON

- **We define the likelihood function:**

$$L(x_1, x_2, \cdots, x_n; \theta) = P(X_1 = x_1, X_2 = x_2, \cdots, X_n = x_n; \theta)$$

- **So for our model:**

$$(\alpha_i, \beta_i,\ i = 1, \ldots, n) = \prod_{k=1}^{N} \frac{e^{-\lambda_k} \lambda_k^{x_k}}{x_k!} \frac{e^{-\mu_k} \mu}{y_k!} \quad , \qquad \theta = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \\ \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}$$

N nombre de matchs du dataset, k représente le kème match

- **We redefine λ, μ by expressing their logarithm as a linear combination of the variables:**

$$\ln(\lambda_k) = \alpha_{i(k)} + \beta_{j(k)} \qquad \ln(\mu_k) = \alpha_{j(k)} + \beta_{i(k)}$$

- **•Objective determine α, β for each team to maximize L, thus the likelihood between the data and the model**

POISSON REGRESSION

•B- ESTIMATION OF MAXIMUM LIKELIHOOD

- **We work with the logarithm of L to facilitate the calculations:**

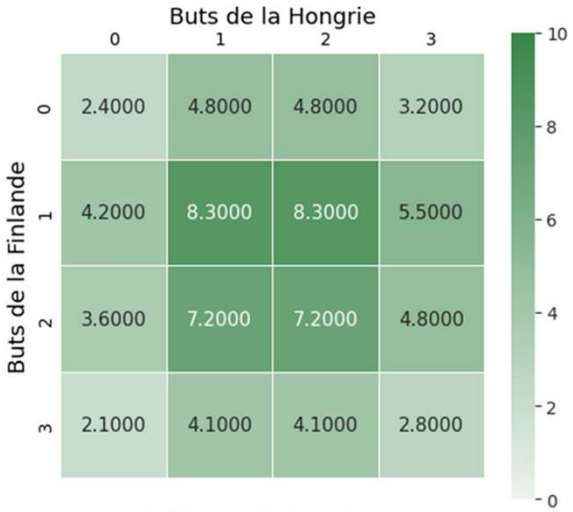
$$l(\alpha_i, \beta_i,\ i = 1, \ldots, n) = ln(L(\alpha_i, \beta_i,\ i = 1, \ldots, n))$$
$$= (\sum_{i=1}^{N} \ln(\frac{1}{x_k!}) - \lambda_k + x_k \ln(\lambda_k)) + (\sum_{i=1}^{N} \ln(\frac{1}{y_k!}) - \mu_k + y_k \ln(\mu_k))$$

- **To minimize the need to cancel out its gradient, either find the optimal parameters such as:**

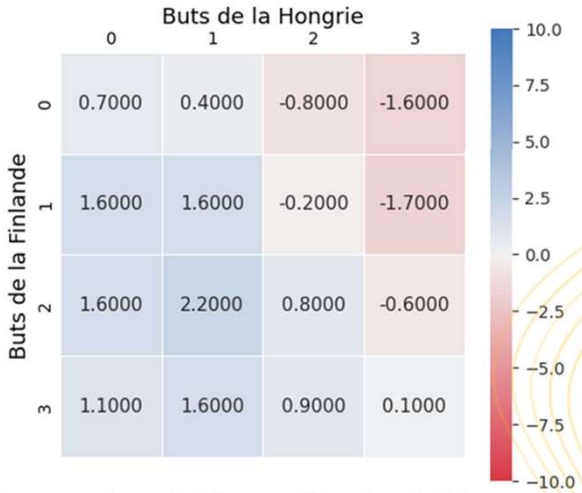
$$\nabla l(\alpha_i, \beta_i,\ i = 1, \ldots, n) = \begin{bmatrix} \frac{\partial l}{\partial \alpha_1}(\alpha_i, \beta_i,\ i = 1, \ldots, n) \\ \frac{\partial l}{\partial \alpha_2}(\alpha_i, \beta_i,\ i = 1, \ldots, n) \\ \vdots \\ \frac{\partial l}{\partial \beta_n}(\alpha_i, \beta_i,\ i = 1, \ldots, n) \end{bmatrix} = \vec{0}$$

POISSON REGRESSION

•RESULT: 2ND PREDICTION FROM FINLAND – HUNGARY AND DIFFERENCE WITH THE 1ST MODEL



Probabilités: Finland - Hungary



Différence de prévisions : Simple et Régression

CONCLUSION

34%

- **Finland wins**

21%

- **Draw**

45%

- **Hungary wins**

POISSON REGRESSION

•RELEVANCE OF THE IMPROVEMENT OF THE MODEL: COMPARISON OF THE RESULTS TO THE BOOKMAKERS' PREDICTIONS

1st model

Team 1 victory
17% difference

Equality
9% difference

Team 2 Victory
16% difference

2nd model

Team 1 victory
11%% difference

Equality
7% difference

Team 2 Victory
9% difference

=> On remarque une amélioration notable de la précision

03 - ADDITION OF AN ADDITIONAL VARIABLE: FIFA RANKING

A) MODEL ADJUSTMENT

CL		Équipe	Total points
1		Argentine	1858
2		France	1840.59
3	↑ 1	Belgique	1795.23
4	↓ 1	Angleterre	1794.9
5		Brésil	1788.65
6	↑ 1	Portugal	1748.11
7	↓ 1	Pays-Bas	1742.29
8		Espagne	1727.5

Classement Fifa des équipes internationales

	HomeCountry	Home_Team_Fifa_Points
0	San Marino	753.11
1	Gibraltar	851.64
2	Faroe Islands	1126.30
3	Latvia	1096.19
4	Lithuania	1069.97
5	Azerbaijan	1143.22
6	Kazakhstan	1198.24
7	Kosovo	1179.69
8	Estonia	1179.30
9	Cyprus	1163.73
10	Belarus	1212.28

Mise en forme des données pour l'exploitation

ADDITION OF AN ADDITIONAL VARIABLEFIFA RANKING

- •We transform the expression of λ and μ

$$ln(\lambda) = \alpha_i + \beta_j + \gamma \cdot C.Fifa_i + \delta \cdot C.Fifa$$

$$ln(\mu) = \alpha_j + \beta_i + \gamma \cdot C.Fifa_i + \delta \cdot C.Fifa$$

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-0.7620	0.444	-1.718	0.086	-1.632	0.107
homePoints	0.0021	0.000	9.228	0.000	0.002	0.003
awayPoints	-0.0014	0.000	-7.253	0.000	-0.002	-0.001

CONSTRUCTION OF A SIMPLE FIRST MODEL

- **WE USE THE STRATSMODEL GLM MODULE WHICH RETURNS THE OPTIMAL PARAMETERS THAT MINIMIZE LOG-LIKELIHOOD:**

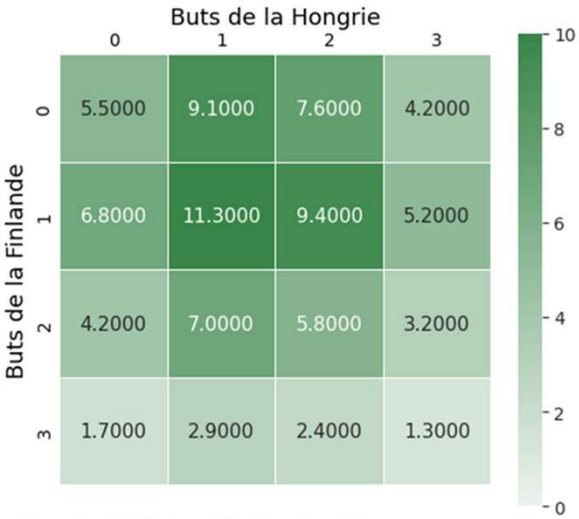
Generalized Linear Model Regression Results

Dep. Variable:	goals	**No. Observations:**	239
Model:	GLM	**Df Residuals:**	186
Model Family:	Poisson	**Df Model:**	52
Link Function:	Log	**Scale:**	1.0000
Method:	IRLS	**Log-Likelihood:**	-344.46
Date:	Sun, 02 Jun 2024	**Deviance:**	238.43
Time:	12:23:37	**Pearson chi2:**	214.
No. Iterations:	23	**Pseudo R-squ. (CS):**	0.5060
Covariance Type:	nonrobust		

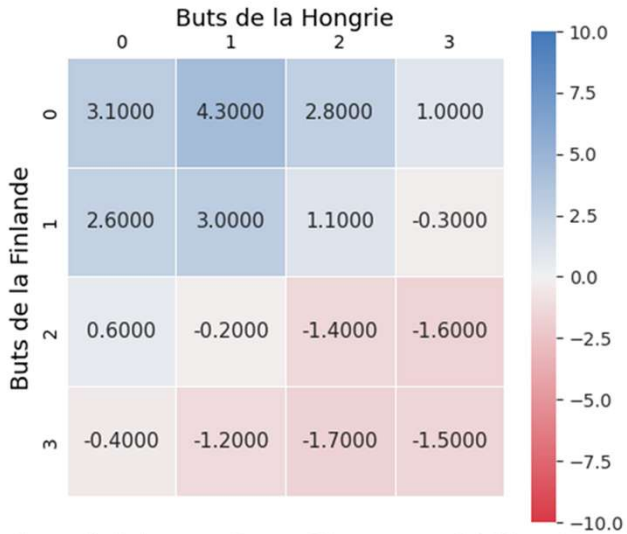
	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	0.5596	0.378	1.481	0.139	-0.181	1.300
team[T.Andorra]	-2.1691	1.069	-2.029	0.042	-4.264	-0.074
team[T.Armenia]	-0.5596	0.627	-0.893	0.372	-1.788	0.669
team[T.Austria]	0.3567	0.493	0.724	0.469	-0.609	1.323
team[T.Azerbaijan]	-0.5596	0.627	-0.893	0.372	-1.788	0.669
team[T.Belarus]	-0.7828	0.627	-1.249	0.212	-2.011	0.446
team[T.Belgium]	0.5390	0.476	1.133	0.257	-0.393	1.471
team[T.Bosnia-Herzegovina]	-0.4055	0.535	-0.759	0.448	-1.453	0.642
team[T.Bulgaria]	-0.8473	0.690	-1.228	0.220	-2.200	0.505

ADDITION OF AN ADDITIONAL VARIABLE-FIFA RANKING

•RESULT: 3RD PREDICTION FROM FINLAND – HUNGARY AND DIFFERENCE WITH THE 2ND MODEL



Probabilités: Finland - Hungary en %



Différence de prévisions : Sans Classement Fifa et avec en %

CONCLUSION

28%
- Finland wins

24%
- Draw

48%
- Hungary wins

ADDITION OF AN ADDITIONAL VARIABLEFIFA RANKING

RELEVANCE OF THE IMPROVEMENT OF THE MODEL: COMPARISON OF THE RESULTS TO THE BOOKMAKERS' PREDICTIONS

1st model

Team 1 victory
17% difference

Equality
9% difference

Team 2 Victory
16% difference

1st model

Team 1 victory
11% difference

Equality
7% difference

Team 2 Victory
9% difference

1st model

Team 1 victory
5% difference

Equality
7% difference

Team 2 Victory
5% difference

Conclusion: Le modèle a été encore amélioré, le 3ème à jusqu'à 3x moins de différences que le premier

04- COMPARISON TO REAL DATA: EURO 2020

- **Interest:**
- **Tournament over -> We can compare model and reality**
- **More data to assess the strength of the teams at the time of the tournament, difficulty specific to the Euro**

Bracket [edit]

Round of 16		Quarter-finals		Semi-finals		Final	
27 June 2021 – Seville							
Belgium	1	2 July 2021 – Munich					
Portugal	0	Belgium	1				
26 June 2021 – London		**Italy**	2				
Italy (a.e.t.)	2			6 July 2021 – London			
Austria	1			**Italy** (p)	1 (4)		
28 June 2021 – Bucharest				Spain	1 (2)		
France	3 (4)	2 July 2021 – Saint Petersburg					
Switzerland (p)	3 (5)	Switzerland	1 (1)				
28 June 2021 – Copenhagen		**Spain** (p)	1 (3)				
Croatia	3					11 July 2021 – London	
Spain (a.e.t.)	5					**Italy** (p)	1 (3)
29 June 2021 – Glasgow						England	1 (2)
Sweden	1	3 July 2021 – Rome					
Ukraine (a.e.t.)	2	Ukraine	0				
29 June 2021 – London		**England**	4				
England	2			7 July 2021 – London			
Germany	0			**England** (a.e.t.)	2		
27 June 2021 – Budapest				Denmark	1		
Netherlands	0	3 July 2021 – Baku					
Czech Republic	2	Czech Republic	1				
26 June 2021 – Amsterdam		**Denmark**	2				
Wales	0						
Denmark	4						

COMPARISON TO ACTUAL DATA: EURO 2020

Remarks:

11 matches correctly predicted for 4 matches incorrect >75% of matches correctly predicted

• Reflects the reality > Interesting for sports betting

Incorrectly predicted matches are either very close or big surprises

Matchs correctement prédis

Matchs incorrectement prédis

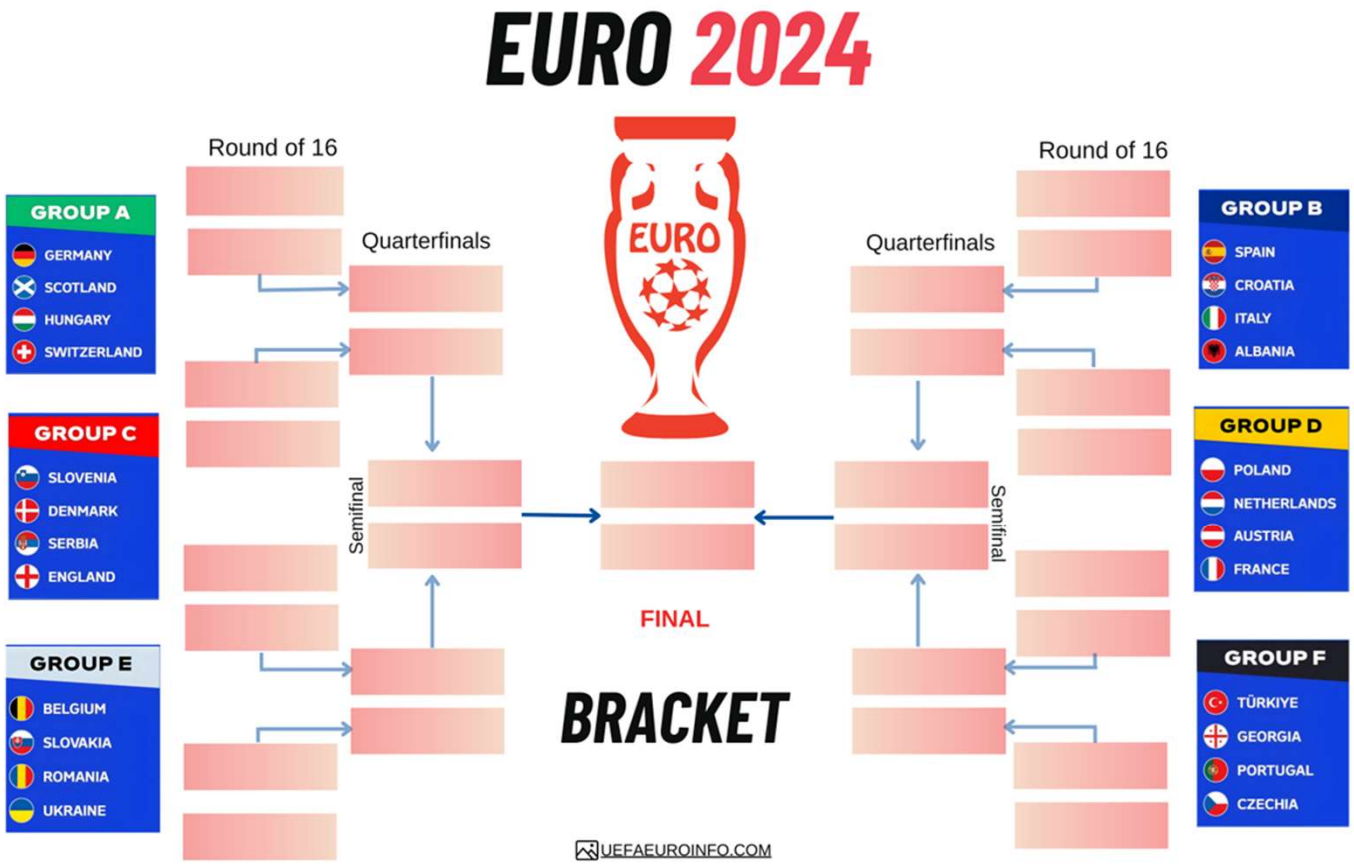
Bracket [edit]

Round of 16		Quarter-finals		Semi-finals		Final	
27 June 2021 – Seville							
Belgium	1	2 July 2021 – Munich					
Portugal	0	Belgium	1				
26 June 2021 – London		**Italy**	2	6 July 2021 – London			
Italy (a.e.t.)	2			**Italy** (p)	1 (4)		
Austria	1			Spain	1 (2)		
28 June 2021 – Bucharest						11 July 2021 – London	
France	3 (4)	2 July 2021 – Saint Petersburg				**Italy** (p)	1 (3)
Switzerland (p)	3 (5)	Switzerland	1 (1)			England	1 (2)
28 June 2021 – Copenhagen		**Spain** (p)	1 (3)				
Croatia	3						
Spain (a.e.t.)	5						
29 June 2021 – Glasgow							
Sweden	1	3 July 2021 – Rome					
Ukraine (a.e.t.)	2	Ukraine	0				
29 June 2021 – London		**England**	4	7 July 2021 – London			
England	2			**England** (a.e.t.)	2		
Germany	0			Denmark	1		
27 June 2021 – Budapest							
Netherlands	0	3 July 2021 – Baku					
Czech Republic	2	Czech Republic	1				
26 June 2021 – Amsterdam		**Denmark**	2				
Wales	0						
Denmark	4						

05- PREDICTION OF FUTURE RESULTS: APPLICATION TO EURO 2024

A) Competition

- 6 groups, then a knockout tournament
- Germany plays at home
- Advantage given to Germany: increase in its Fifa ranking by 150 points



05- PREDICTION OF FUTURE RESULTS: APPLICATION TO EURO 2024

B) Euro 2024 final tree

Impossible to know the matches in advance

because of the tournament system

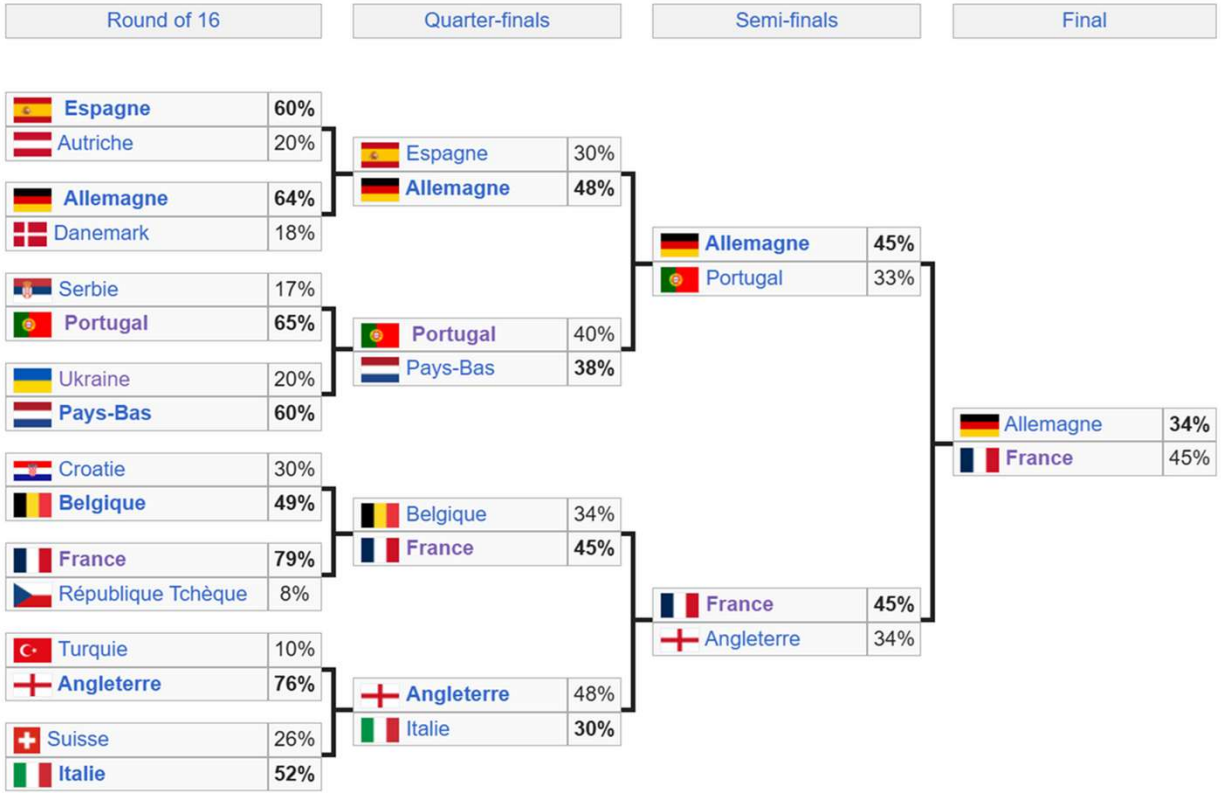
We choose the best 3rd graders and we place

them at random

Germany is doing very well thanks to the home

boost

Victory for France!



CONCLUSION AND DISCUSSION

IS VERY CLOSE TO REALITY BUT REMAINS A PREDICTION

HYPOTHESIS SCORE INDEPENDENCE = STRONG HYPOTHESIS
-->Bivariate Poisson with corrective term that binds the two teams

-TIME WEIGHTING: TAKING INTO ACCOUNT TIME IN A MATCH

-ADDITION OF SEVERAL OTHER POTENTIAL VARIABLES: GDP OF A COUNTRY, MARKET VALUE OF THE TEAM, NUMBER OF PLAYERS WHO PLAY IN THE CHAMPIONS LEAGUE...

VI- ANNEX

A) DATA

```
import pandas as pd
données=pd.read_csv('/content/drive/MyDrive/CSV/international-uefa-euro-qualifiers-matches-2022-to-2024-stats (1).csv')
données=données[['date_GMT','home_team_name','away_team_name','home_team_goal_count','away_team_goal_count','total_goal_count']]
données.head()
données_propres=données[['home_team_name','away_team_name','home_team_goal_count','away_team_goal_count']]
epl_1718 = données_propres.rename(columns={'home_team_goal_count': 'HomeGoals', 'away_team_goal_count': 'AwayGoals','home_team_name': 'Ho
epl_1718
```

```
[63] merged1 = pd.merge(fifarank_Home, epl_1718, left_on='HomeCountry', right_on='HomeTeam',how='right',indicator='Exists')
     merged2 = pd.merge(fifarank_Away, merged1, left_on='AwayCountry', right_on='AwayTeam',how='right',indicator=True)
     tableau=merged2[['HomeTeam','AwayTeam','HomeGoals','AwayGoals','Home_Team_Fifa_Points','Away_Team_Fifa_Points']]
     tableau
```

	HomeTeam	AwayTeam	HomeGoals	AwayGoals	Home_Team_Fifa_Points	Away_Team_Fifa_Points
0	Kazakhstan	Slovenia	1	2	1198.24	1391.04
1	Italy	England	1	2	1726.58	1797.39
2	FYR Macedonia	Malta	2	1	1350.53	972.86
3	Denmark	Finland	3	1	1597.37	1419.47
4	San Marino	Northern Ireland	0	2	753.11	1361.17
...	...	...	...	...	...	...
234	Israel	Iceland	1	4	1323.81	1352.98
235	Bosnia-Herzegovina	Ukraine	1	2	1381.10	1541.25
236	Georgia	Greece	0	0	1312.80	1441.06
237	Wales	Poland	0	0	1506.04	1536.99
238	Ukraine	Iceland	2	1	1541.25	1352.98

239 rows × 6 columns

VI- ANNEX

B) SIMPLE POISSON

```python
def prediction_match(domicile,exterieur):
  lambda_domicile=données_forces_equipes.at[domicile,'Buts marqués'] * données_forces_equipes.at[exterieur,'Buts concédés']
  lambda_exterieur=données_forces_equipes.at[exterieur,'Buts marqués'] * données_forces_equipes.at[domicile,'Buts concédés']
  Tableau_proba=[]
  for i in range(0,20):
      for j in range(0,20):
          proba1 = poisson.pmf(i,lambda_domicile)
          proba2 = poisson.pmf(j,lambda_exterieur)
          proba_match=proba1*proba2
          Tableau_proba.append([i,j,proba_match])
  return Tableau_proba
```

```python
[70] cmap = sns.diverging_palette(10, 133, as_cmap=True)
     fig, ax = plt.subplots(figsize=(5,5))
     with sns.axes_style("white"):
         ax = sns.heatmap(arrange_tableau_rg("Finland","Hungary"),
                          annot=True, fmt='.4f', cmap=cmap, vmin=0, vmax=10, center=0.00,
                          square=True, linewidths=.5, annot_kws={"size": 11}, cbar_kws={"shrink": .8})
         ax.tick_params(axis=u'both', which=u'both',length=0)
         ax.grid(which='major', axis='both', linestyle='')
         ax.set_xlabel('Buts de la Hongrie', fontsize=13)
         ax.set_ylabel('Buts de la Finlande', fontsize=13)
         ax.xaxis.set_label_position('top')
         ax.xaxis.set_ticks_position('top')
         plt.figtext(0.45, 0.07, '  Probabilités: Finland - Hungary', horizontalalignment='center',
                     fontsize=14, multialignment='left', fontweight='bold')
     plt.tight_layout()
     plt.show()
```

```python
prediction_match('France','Netherlands')
```

```
[[0, 0, 0.018897039677585683],
 [0, 1, 0.015058578493076092],
 [0, 2, 0.005999902368335005],
 [0, 3, 0.001593724066588986],
 [0, 4, 0.00031749971639077445],
 [1, 0, 0.05993904772734209],
 [1, 1, 0.04776392865772573],
 [1, 2, 0.019030940324562595],
 [1, 3, 0.00505509352371194],
 [1, 4, 0.0010070694129269877],
 [2, 0, 0.09505958350508158],
 [2, 1, 0.07575060560561188],
 [2, 2, 0.030181881920985986],
 [2, 3, 0.008017062385261902],
 [2, 4, 0.0015971491470638942],
 [3, 0, 0.10050570547672685],
 [3, 1, 0.08009048405176672],
 [3, 2, 0.0319110522393758],
 [3, 3, 0.008476373251084197],
 [3, 4, 0.0016886524836144294],
 [4, 0, 0.07969788363974825],
 [4, 1, 0.06350925102542439],
 [4, 2, 0.025304467205442528],
 [4, 3, 0.006721499101445672],
 [4, 4, 0.0013390486491161296]]
```

VI- ANNEX

C) POISSON REGRESSION: GLM

```python
# importing the tools required for the Poisson regression model
import statsmodels.api as sm
import statsmodels.formula.api as smf

goal_model_data = pd.concat([epl_1718[['HomeTeam','AwayTeam','HomeGoals']].assign(home=1).rename(
            columns={'HomeTeam':'team', 'AwayTeam':'opponent','HomeGoals':'goals'}),
           epl_1718[['AwayTeam','HomeTeam','AwayGoals']].assign(home=0).rename(
            columns={'AwayTeam':'team', 'HomeTeam':'opponent','AwayGoals':'goals'})])

goal_model_data


poisson_model_rg = smf.glm(formula="goals ~ team + opponent", data=goal_model_data,
                        family=sm.families.Poisson()).fit()
poisson_model_rg.summary()
```

```python
# importing the tools required for the Poisson regression model
import statsmodels.api as sm
import statsmodels.formula.api as smf

goal_model_data = tableau[['HomeTeam','AwayTeam','HomeGoals','Home_Team_Fifa_Points','Away_Team_Fifa_Points']].rename(
            columns={'HomeTeam':'team', 'AwayTeam':'opponent','HomeGoals':'goals','Home_Team_Fifa_Points':'homePoints','Away_Team_Fifa_Points':'awayPoints'})
goal_model_data


poisson_model_fk = smf.glm(formula="goals ~ team + opponent + homePoints + awayPoints", data=goal_model_data,
                        family=sm.families.Poisson()).fit()
poisson_model_fk.summary()
goal_model_data
```

```python
def simulate_match_rg(foot_model, homeTeam, awayTeam, max_goals=10):
    home_goals_avg = foot_model.predict(pd.DataFrame(data={'team': homeTeam,
                                                           'homePoints': points(homeTeam), 'opponent':awayTeam, 'awayPoints': points(awayTeam)},
                                                     index=[1])).values[0]
    away_goals_avg = foot_model.predict(pd.DataFrame(data={'team': awayTeam,
                                                           'homePoints': points(awayTeam),'opponent':homeTeam, 'awayPoints': points(homeTeam)},
                                                     index=[1])).values[0]
    team_pred = [[poisson.pmf(i, team_avg) for i in range(0, max_goals+1)] for team_avg in [home_goals_avg, away_goals_avg]]
    return(np.outer(np.array(team_pred[0]), np.array(team_pred[1])))
simulate_match_rg(poisson_model_rg, 'Turkey', 'Greece', max_goals=5)
```

VI- ANNEX

D) POISSON BY HAND REGRESSION

```python
def dc_log_like(x, y, alpha_x, beta_x, alpha_y, beta_y, rho, gamma):
    lambda_x, mu_y = np.exp(alpha_x + beta_y), np.exp(alpha_y + beta_x)
    return (np.log(poisson.pmf(x, lambda_x)) + np.log(poisson.pmf(y, mu_y)))

def solve_parameters(dataset, debug = False, init_vals=None, options={'disp': True, 'maxiter':100},
                     constraints = [{'type':'eq', 'fun': lambda x: sum(x[:20])-20}] , **kwargs):
    teams = np.sort(dataset['HomeTeam'].unique())

    if init_vals is None:
        # Initialisation aléatoire des paramètres
        init_vals = np.concatenate((np.random.uniform(0,1,(n_teams)), # Force d'attaque
                                    np.random.uniform(0,-1,(n_teams)), # Force de défense
                                   ))
    print(init_vals)
    def dc_log_like(x, y, alpha_x, beta_x, alpha_y, beta_y):
        lambda_x, mu_y = np.exp(alpha_x + beta_y), np.exp(alpha_y + beta_x)
        return (np.log(poisson.pmf(x, lambda_x)) + np.log(poisson.pmf(y, mu_y)))

    def estimate_paramters(params):
        score_coefs = dict(zip(teams, params[:n_teams]))
        defend_coefs = dict(zip(teams, params[n_teams:(2*n_teams)]))
        log_like = [dc_log_like(row.HomeGoals, row.AwayGoals, score_coefs[row.HomeTeam], defend_coefs[row.HomeTeam],
                    score_coefs[row.AwayTeam], defend_coefs[row.AwayTeam] for row in dataset.itertuples()]
        print(-sum(log_like))
        return -sum(log_like)

    opt_output = minimize(estimate_paramters, init_vals, options=options, constraints = constraints, **kwargs)


    return dict(zip(["attack_"+team for team in teams] +
                    ["defence_"+team for team in teams]],

                    opt_output.x))

solve_parameters(tableau)
```

```
params={'attack_Albania': 1.5588989499844084,
 'attack_Andorra': -0.7132454018825906,
 'attack_Armenia': 1.3271054230595076,
 'attack_Austria': 1.1930419246678428,
 'attack_Azerbaijan': 0.4722606319203112,
 'attack_Belarus': 0.3117175637344285,
 'attack_Belgium': 1.397255088778856,
 'attack_Bosnia-Herzegovina': 1.1324348317446344,
 'attack_Bulgaria': 0.4475932795356825,
 'attack_Croatia': 1.53442478024527,
 'attack_Cyprus': 0.9628272337769449,
 'attack_Czech Republic': 1.6228018919948033,
 'attack_Denmark': 0.4679698104094707,
 'attack_England': 2.1358255079384967,
 'attack_Estonia': -0.3690096603718759,
 'attack_FYR Macedonia': 1.6786784022123384,
 'attack_Faroe Islands': -0.03602861606939084,
 'attack_Finland': 0.45169926814317973,
 'attack_France': 2.433027919574684,
 'attack_Georgia': 1.9907211376789875,
 'attack_Gibraltar': -12.092967744446211,
 'attack_Greece': 1.6535386014413638,
 'attack_Hungary': 1.1553115657306459,
 'attack_Iceland': 1.7203243541657631,
 'attack_Israel': 0.5084120556228477,
 'attack_Italy': 1.913165051675244,
 'attack_Kazakhstan': 0.3029711690287984,
 'attack_Kosovo': 0.35299456278842534,
 'attack_Latvia': 0.9267159243528204,
 'attack_Liechtenstein': -1.0373803886318187,
 'attack_Lithuania': 0.5887533628653139,
 'attack_Luxembourg': 1.3736698208645362,
 'attack_Malta': -0.01603622838947939,
 'attack_Moldova': 1.1761897359161377,
 'attack_Montenegro': 0.6419964725930145,
 'attack_Netherlands': 1.9624011897492564,
 'attack_Northern Ireland': -0.2590082095684746,
 'attack_Norway': 2.2583444817192677,
 'attack_Poland': 1.491606221415408,
 'attack_Portugal': 2.263851547881049,
 'attack_Republic of Ireland': 1.360588773793498,
 'attack_Romania': 0.7531094788930679,
 'attack_San Marino': -1.1206504856202144,
```

VI- ANNEX

D) BETMARKET COMPARISON

```python
def get_difference_reg():
  erreur_stat=[0,0,0]
  for i in range(0,22):
    homeTeam=betodds2['Team1'][i]
    awayTeam=betodds2['Team2'][i]
    index=i
    betPredictions=[betodds2['prob1'][i],betodds2['probX'][i],betodds2['prob2'][i]]
    regPredictions=prob_resultat(homeTeam, awayTeam)
    erreur_stat=somme_listes(erreur_stat, diff_list(betPredictions,regPredictions))
  return divise_liste(erreur_stat, 22)

def get_difference_reg_fk():
  erreur_stat=[0,0,0]
  for i in range(0,22):
    homeTeam=betodds2['Team1'][i]
    awayTeam=betodds2['Team2'][i]
    index=i
    betPredictions=[betodds2['prob1'][i],betodds2['probX'][i],betodds2['prob2'][i]]
    regPredictions=prob_resultat_fk(homeTeam, awayTeam)
    erreur_stat=somme_listes(erreur_stat, diff_list(betPredictions,regPredictions))
  return divise_liste(erreur_stat, 22)

def get_difference_sim():
  erreur_stat=[0,0,0]
  for i in range(0,22):
    homeTeam=betodds2['Team1'][i]
    awayTeam=betodds2['Team2'][i]
    index=i
    betPredictions=[betodds2['prob1'][i],betodds2['probX'][i],betodds2['prob2'][i]]
    simPredictions=probabilités_simple(homeTeam, awayTeam)
    erreur_stat=somme_listes(erreur_stat, diff_list(betPredictions,simPredictions))
  return divise_liste(erreur_stat, 22)

(get_difference_sim(),get_difference_reg(),get_difference_reg_fk())
```

```
[0.17661619661981146, 0.09373468875687632, 0.16302261046884198],
[0.11140319646405751, 0.07110864012864794, 0.09508869770037474],
[0.05616073847402321, 0.07164051189382267, 0.05717392413647801])
```